Final Project Design Phase

Computer Science - Week 11

Oct 3, 2026 - Version 0.0.4

For complete instructions and project ideas, refer to the assignment in Canvas. Specifically, review the final project requirements before you continue below. If you have not reviewed the project requirements in Canvas (you might have to go to the modules view and check the current or past week's content), you will not be able to properly complete the parts below. In addition, it may be helpful to review the information on this Drafter documentation page.

0) Project Planning and Preparation

Successful programmers spend time planning before they begin coding. *After* reviewing the project requirements, answer the following questions to help guide your design and development process.

A. What concepts from class or previous assignments might help you complete this project? Which programming concepts (loops, conditionals, lists, classes, etc.) do you expect to use? What programming skills do you want to improve while completing this project?

I think I will need to use all parts of this class to create my program. I want to improve my efficiency in programming. I would like to be able to code faster and more accurately than I do currently.

B. Which parts of the project do you expect to be the easiest for you? The hardest? Why?

The actual calculations, the base program, of the project should be fairly easy. Stoichiometry is not very difficult. However, importing the chemistry library and implementing Drafter will be very difficult because I have never had to work with multiple files inside of my code.

C. Even though you have not yet designed the final project, estimate which tasks will likely require the most time and effort? Which features or components should be implemented first?

Writing my main program should be quick, but updating the variables to read the data from the chemistry library will take lots of time. Because of this, I should implement the library and familiarize myself with how the data is stored first so that I can build my program around it. Drafter will be implemented last, mainly by the LLM.

D. What distractions might interfere with your work and how will you minimize them? What resources will you need available while working? What questions, if any, do you still have about the assignment requirements?

Other classes may be distracting because I have other work to do for them, but I will manage my time so as to dedicate the majority of my time to this project. I will need ChatGPT, the BlockPy textbook, Stack Overflow, and the CISC Discord. I have no more questions yet.

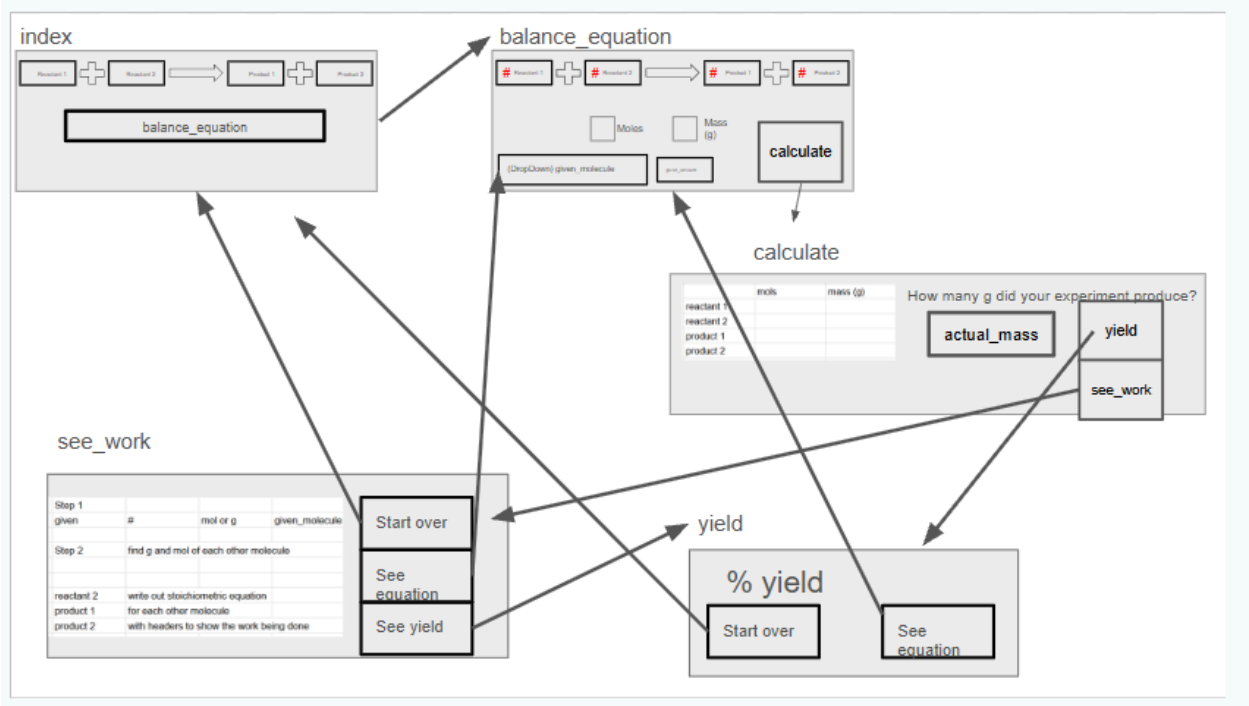
1) Site Diagram

Create a mock-up of your planned website. We suggest you make the mockup on paper, although you can also use software like PowerPoint or Slides to make a mockup. There is other design software that could also be used, but we strongly encourage you to keep it simple. Do not spend more than 15-30 minutes detailing the mock-up. Your goal should be to make a very quick and dirty diagram, to show that you have some kind of idea.

We recommend drawing each page as a rectangle, and laying out the text, buttons, and other components inside. Then, use arrows to indicate links between pages. Label each page the way you will name each function, and then put labels on the arrows to indicate what the button links. Also write down any parameters that should come into the pages.

In addition to the pages and routes, write down what fields your `State` will have, both their names and types. Think clearly about what data will need to be persistent across the entire website.

Add photo(s) or screenshot(s) of your website mock-up and labeled diagram on the next page.


index
balance_equation
balance_equation
balance_equation
Moles
Mass (g)
calculate
calculate
mols
mass (g)
reactant 1
reactant 2
product 1
product 2
How many g did your experiment produce?
actual_mass
yield
see_work
see_work
Step 1
given
#
mol or g
given_molecule
Step 2
find g and mol of each other molecule
reactant 2
write out stoichiometric equation
product 1
for each other molecule
product 2
with headers to show the work being done
Start over
See equation
See yield
yield
% yield
Start over
See equation

2) Implementing Requirements

This table should serve as a draft for how you will implement the requirements into your final project. You may change your ideas later on, but your final site must fulfill all the requirements.

Requirement	**Implementation**
Purpose/Functionality A 1-2 sentence description of the purpose or functionality of your site	I am going to make a program that addresses the suggested Chemical Engineering goal. It will take in an unbalanced equation, balance it, then calculate unknown masses/moles based on an input and also calculate percent yield and whether or not the reaction was exothermic or endothermic.
State Dataclass Attributes Minimum of 4 attributes, one of which must be a list/dict of instances of your second dataclass. Briefly describe each attribute, its type, and how it will be meaningfully modified throughout the site.	Equation: list[Molecule] In order, the list will represent the first reactant, second reactant, first product, second product. It will be modified on the first page to be given elements. Coefficients: list[int] This list represents the stoichiometric coefficients following the same order and modifications as the equation attribute. Actual_mass: int Represents the mass the user found in their experiment. Will be given a value on the calculate page. Moles: bool Mass: bool These are checkboxes found on the balance_equation page. They will be given value on the page and will be used to determine the contents of the calculate page.
Second Dataclass Your second dataclass, including	Molecule Mass: float

its attributes, their types, and how they will be meaningfully modified (if they are modified)	Symbol: str These attributes will not be modified
Routes (minimum 7) The names of your route functions	Index Balance_equation Calculate Yield See_work Start_over see_equation
Pages (minimum 5) A brief description of the pages in your site	Index: allows the user to input a an unbalanced equation in textboxes then allows the user to press a button to balance the equation Balance_equation: shows the balanced equation. Allows the user to choose what information they are working with (known molecule and amounts), then allows the user to calculate the theoretical yield Calculate: shows all moles and grams for the reaction to go to completion and the amount of product produced if this were the case. Has buttons to calculate percent yield if you input the actual product produced in your experiment Yield: shows the percent yield of your experiment with buttons to go back to previous pages. See_work: shows a step by step on how the program found the numbers it did with buttons to go back to previous pages.
User Input (minimum 3) Description of user input	4 Textboxes for inputting the elements in your reaction 2 check boxes for determining whether you want

components (TextBox, TextArea, CheckBox, or SelectBox) and their purposes For more information on these input types, see Drafter Documentation.	answers in moles or grams 1 select box determining which molecule you have a known quantity of.
If Statements (minimum 3) Brief description of what will be evaluated in your if statements and how they will meaningfully change the logic of the site	1. To check if the element symbols inputted are actually real elements. If not, the website will require a re-input. 2. To determine which calculation pathway to go through whether the user chose moles or mass. 3. To prevent the program from dividing by zero in the yield page.
Loop (minimum 1) This loop must iterate over the list/dict of dataclass instances in your State attribute. Description of this loop and why it is meaningful to the site	For each molecule in the equation attribute, the program will need to read its name and molar mass to be used in the stoichiometric equation.

LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

I did not use an LLM on this assignment

Final Submission

When you are finished, download this file as a Word Document (docx) and upload the file to the appropriate assignment on Canvas.